PHYDSC401T — Mathematical Physics-III

Unit 2: Partial Differential Equations and Special Functions

7th Semester, Assam University (FYUG) — Haflong Government College

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Notes 5 of 20 — drafted by gemini-3.6-flash (Kaggle Model Proxy) from the unit syllabus only, without external sources, and checked against the syllabus.

Index

Unit syllabus (as in the syllabus file)

1. Solutions of partial differential equations using separation of variables: Laplace's equation in problems of rectangular, cylindrical and spherical symmetry.
2. Wave equation and its solution for vibrational modes of a stretched string, rectangular and circular membranes.
3. Diffusion equation.
4. Hermite polynomials and recurrence relation.
5. Laguerre polynomials.

Top 2 high-mark topics in this unit

1. Laplace equation by separation of variables in rectangular, cylindrical and spherical symmetry — named first in the syllabus for this unit and the main separation-of-variables derivation, a standard long-answer topic.
2. Wave equation and vibrational modes of a stretched string and of rectangular and circular membranes — the other named long derivation in this unit (wave equation plus modes), with a 7-mark derivation in the Mumbai paper.

Contents

1. Separation of Variables: The Method
2. Laplace's Equation in Rectangular Coordinates
3. Laplace's Equation in Cylindrical and Spherical Coordinates
4. The Wave Equation and Vibrational Modes of a Stretched String
5. Vibrational Modes of Rectangular and Circular Membranes
6. The Diffusion Equation
7. Hermite Polynomials and Recurrence Relations
8. Generating Function of Hermite Polynomials
9. Laguerre Polynomials and Recurrence Relations
10. Questions — Short form, then Long form (no AU questions found for this unit)
11. Sources

1. Separation of Variables: The Method

The method of separation of variables is a powerful analytical technique used to solve linear homogeneous partial differential equations (PDEs) subject to prescribed boundary and initial conditions. The fundamental assumption of this method is that a function of multiple independent variables can be expressed as a product of several single-variable functions, each depending on only one independent variable.

Consider a second-order linear homogeneous partial differential equation for an unknown field $u(x, y)$ depending on two spatial variables x and y. We postulate that the solution can be written in the trial product form

$$u(x, y) = X(x)Y(y)$$

where $X(x)$ is a function purely of x, and $Y(y)$ is a function purely of y.

To see how this simplifies the differential equation, we calculate the partial derivatives of $u(x, y)$ with respect to x and y. The partial derivatives become ordinary derivatives of the single-variable constituent functions. The first-order partial derivatives are given by

$$\frac{\partial u}{\partial x} = \frac{\mathrm{d}X(x)}{\mathrm{d}x}Y(y) = X'(x)Y(y)$$
$$\frac{\partial u}{\partial y} = X(x)\frac{\mathrm{d}Y(y)}{\mathrm{d}y} = X(x)Y'(y)$$

Similarly, the second-order partial derivatives are expressed as

$$\frac{\partial^2 u}{\partial x^2} = X''(x)Y(y)$$
$$\frac{\partial^2 u}{\partial y^2} = X(x)Y''(y)$$

Substituting these partial derivatives into a governing partial differential equation allows us to separate the variables. For a general linear spatial operator equation such as $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$, substitution of the product solution yields

$$X''(x)Y(y) + X(x)Y''(y) = 0$$

Dividing the entire equation by the non-zero product function $u(x, y) = X(x)Y(y)$ isolates the functional dependencies on opposite sides of the equation:

$$\frac{X''(x)}{X(x)} + \frac{Y''(y)}{Y(y)} = 0$$

Rearranging the terms puts all functions of x on the left-hand side and all functions of y on the right-hand side:

$$\frac{X''(x)}{X(x)} = -\frac{Y''(y)}{Y(y)}$$

The left-hand side of this equation depends exclusively on x, whereas the right-hand side depends exclusively on y. Because x and y are independent variables, varying x cannot affect the value of the right-hand side, and varying y

cannot affect the value of the left-hand side. Consequently, both sides must equal a common constant, known as the separation constant. Denoting this constant by λ, we obtain two separate ordinary differential equations (ODEs):

$$\frac{X''(x)}{X(x)} = \lambda$$

$$\frac{Y''(y)}{Y(y)} = -\lambda$$

These equations are rewritten as standard linear ordinary differential equations:

$$X''(x) - \lambda X(x) = 0$$

$$Y''(y) + \lambda Y(y) = 0$$

The sign and value of the separation constant λ are determined by the boundary conditions imposed on the physical system. Homogeneous boundary conditions force non-trivial solutions of these ODEs to exist only for a discrete set of allowed values of λ, called eigenvalues. The corresponding solutions $X_n(x)$ and $Y_n(y)$ are the eigenfunctions of the system.

Because the underlying partial differential equation is linear and homogeneous, the principle of superposition applies. The general solution $u(x, y)$ is constructed as a linear combination (an infinite series) of all possible product solutions corresponding to the allowed eigenvalues:

$$u(x, y) = \sum_{n} c_n X_n(x) Y_n(y)$$

where the coefficients c_n are determined using the remaining non-homogeneous boundary condition along with the orthogonality properties of the eigenfunctions.

2. Laplace's Equation in Rectangular Coordinates

Laplace's equation in two-dimensional Cartesian coordinates (x, y) governs steady-state physical phenomena such as electrostatic potential distributions, steady heat conduction, and incompressible fluid potential flow. It is written as

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

To solve Laplace's equation in a rectangular domain, we assume a product solution of the form

$$u(x, y) = X(x)Y(y)$$

Substituting the product solution into Laplace's equation yields

$$X''(x)Y(y) + X(x)Y''(y) = 0$$

Dividing this equation by $X(x)Y(y)$ separates the variables into independent terms:

$$\frac{X''(x)}{X(x)} + \frac{Y''(y)}{Y(y)} = 0$$

Isolating the function of x on one side gives

$$\frac{X''(x)}{X(x)} = -\frac{Y''(y)}{Y(y)}$$

Since x and y vary independently, both sides must equal a single separation constant. To obtain oscillatory solutions in the x-direction that can satisfy zero boundary conditions at two endpoints, we choose a negative separation constant, written as $-k^2$ where k is real. Setting each side equal to $-k^2$ gives

$$\frac{X''(x)}{X(x)} = -k^2$$

$$\frac{Y''(y)}{Y(y)} = k^2$$

This yields two standard ordinary differential equations:

$$X''(x) + k^2 X(x) = 0$$

$$Y''(y) - k^2 Y(y) = 0$$

The general solution for the ordinary differential equation in x is given by trigonometric functions:

$$X(x) = A\cos(kx) + B\sin(kx)$$

The general solution for the ordinary differential equation in y is given by hyperbolic functions:

$$Y(y) = C\cosh(ky) + D\sinh(ky)$$

Thus, a single product solution takes the form

$$u(x,y) = [A\cos(kx) + B\sin(kx)]\,[C\cosh(ky) + D\sinh(ky)]$$

Now consider a rectangular region of width l along the x-axis ($0 \le x \le l$) and height a along the y-axis ($0 \le y \le a$). We consider the boundary conditions where the potential vanishes on three edges of the rectangle, while a prescribed function is applied on the top edge:

$$u(0,y) = 0$$

$$u(l,y) = 0$$

$$u(x,0) = 0$$

$$u(x,a) = \sin\left(\frac{n\pi x}{l}\right)$$

We apply the first boundary condition, $u(0,y) = 0$, to the product solution:

$$X(0)Y(y) = 0 \implies X(0) = 0$$

Evaluating $X(x)$ at $x = 0$ gives

$$X(0) = A\cos(0) + B\sin(0) = A = 0$$

Setting $A = 0$ reduces the spatial eigenfunction in x to

$$X(x) = B\sin(kx)$$

Next, we apply the second boundary condition, $u(l, y) = 0$, which requires

$$X(l)Y(y) = 0 \implies X(l) = 0$$

Evaluating $X(x)$ at $x = l$ gives

$$X(l) = B\sin(kl) = 0$$

For a non-trivial solution ($B \neq 0$), the sine function must vanish, which implies that kl must be an integer multiple of π:

$$kl = m\pi \quad \text{for } m = 1, 2, 3, \ldots$$

This determines the allowed discrete values of the separation constant k, denoted by k_m:

$$k_m = \frac{m\pi}{l}$$

The allowed spatial eigenfunctions in x are therefore

$$X_m(x) = B_m \sin\left(\frac{m\pi x}{l}\right)$$

Now, we apply the third boundary condition at the bottom edge, $u(x, 0) = 0$:

$$X(x)Y(0) = 0 \implies Y(0) = 0$$

Evaluating $Y(y)$ at $y = 0$ using $k_m = \frac{m\pi}{l}$ gives

$$Y(0) = C\cosh(0) + D\sinh(0) = C = 0$$

Setting $C = 0$ reduces the solution for $Y(y)$ to hyperbolic sine functions:

$$Y_m(y) = D_m \sinh\left(k_m y\right) = D_m \sinh\left(\frac{m\pi y}{l}\right)$$

Combining $X_m(x)$ and $Y_m(y)$ into a single product solution $u_m(x, y)$ by defining a unified constant $E_m = B_m D_m$ gives

$$u_m(x, y) = E_m \sin\left(\frac{m\pi x}{l}\right) \sinh\left(\frac{m\pi y}{l}\right)$$

By the principle of superposition, the general solution expressing $u(x, y)$ as a sine-series solution is the sum over all possible mode numbers m:

$$u(x, y) = \sum_{m=1}^{\infty} E_m \sin\left(\frac{m\pi x}{l}\right) \sinh\left(\frac{m\pi y}{l}\right)$$

Finally, we apply the fourth boundary condition at the top edge $y = a$, where $u(x, a) = \sin\left(\frac{n\pi x}{l}\right)$ for a specific integer n:

$$u(x,a) = \sum_{m=1}^{\infty} E_m \sin\left(\frac{m\pi x}{l}\right) \sinh\left(\frac{m\pi a}{l}\right) = \sin\left(\frac{n\pi x}{l}\right)$$

To determine the coefficients E_m, we compare the series expansion term-by-term with the given boundary function. Since the set of functions $\{\sin(m\pi x/l)\}$ forms an orthogonal basis on the interval $[0, l]$, the coefficients must match the single given mode $m = n$.

For all terms with $m \neq n$, the coefficients vanish:

$$E_m = 0 \quad \text{for } m \neq n$$

For the single term $m = n$, equating the coefficients on both sides yields

$$E_n \sinh\left(\frac{n\pi a}{l}\right) = 1$$

Solving for the non-zero coefficient E_n gives

$$E_n = \frac{1}{\sinh\left(\frac{n\pi a}{l}\right)}$$

Substituting E_n back into the general solution collapses the infinite series to a single explicit term corresponding to $m = n$:

$$u(x,y) = \frac{\sin\left(\frac{n\pi x}{l}\right) \sinh\left(\frac{n\pi y}{l}\right)}{\sinh\left(\frac{n\pi a}{l}\right)}$$

This provides the complete and exact solution to Laplace's equation in the given rectangular domain satisfying all specified boundary conditions.

3. Laplace's Equation in Cylindrical and Spherical Coordinates

Laplace's Equation in Cylindrical Coordinates

In three-dimensional space, Laplace's equation for a scalar potential V is given by $\nabla^2 V = 0$. When a physical system possesses cylindrical symmetry, it is natural to express the Laplacian operator in cylindrical coordinates (r, ϕ, z). Assuming independence of the z-coordinate or considering a two-dimensional system in the transverse plane, Laplace's equation in polar coordinates (r, ϕ) takes the form

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial V}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 V}{\partial \phi^2} = 0$$

Expanding the radial derivative using the product rule yields

$$\frac{\partial^2 V}{\partial r^2} + \frac{1}{r}\frac{\partial V}{\partial r} + \frac{1}{r^2}\frac{\partial^2 V}{\partial \phi^2} = 0$$

To solve this partial differential equation, we employ the method of separation of variables by assuming a product form for the potential

$$V(r,\phi) = R(r)\Phi(\phi)$$

Substituting this separable solution into the expanded partial differential equation gives

$$\Phi(\phi)\frac{d^2R}{dr^2} + \frac{\Phi(\phi)}{r}\frac{dR}{dr} + \frac{R(r)}{r^2}\frac{d^2\Phi}{d\phi^2} = 0$$

Multiplying the entire equation by $\frac{r^2}{R(r)\Phi(\phi)}$ separates the variables completely

$$\frac{r^2}{R(r)}\frac{d^2R}{dr^2} + \frac{r}{R(r)}\frac{dR}{dr} + \frac{1}{\Phi(\phi)}\frac{d^2\Phi}{d\phi^2} = 0$$

Rearranging the terms puts the radial dependence on the left-hand side and the angular dependence on the right-hand side

$$\frac{r^2}{R(r)}\frac{d^2R}{dr^2} + \frac{r}{R(r)}\frac{dR}{dr} = -\frac{1}{\Phi(\phi)}\frac{d^2\Phi}{d\phi^2}$$

Since the left-hand side depends solely on r and the right-hand side depends solely on ϕ, both sides must equal a common separation constant, which we denote as m^2. Equating the angular part to m^2 gives

$$-\frac{1}{\Phi(\phi)}\frac{d^2\Phi}{d\phi^2} = m^2$$

Multiplying by $-\Phi(\phi)$ yields the ordinary differential equation for the angular function

$$\frac{d^2\Phi}{d\phi^2} + m^2\Phi(\phi) = 0$$

The general solution to this harmonic oscillator equation is

$$\Phi(\phi) = A_m\cos(m\phi) + B_m\sin(m\phi)$$

Physical boundary conditions in cylindrical coordinates require the potential to be single-valued under a full rotation of 2π, such that $\Phi(\phi + 2\pi) = \Phi(\phi)$. This periodicity condition forces the separation constant m to be a non-negative integer

$$m = 0, 1, 2, 3, \ldots$$

Equating the radial part to m^2 yields

$$\frac{r^2}{R(r)}\frac{d^2R}{dr^2} + \frac{r}{R(r)}\frac{dR}{dr} = m^2$$

Multiplying by $R(r)$ gives the radial Euler-Cauchy equation

$$r^2\frac{d^2R}{dr^2} + r\frac{dR}{dr} - m^2R(r) = 0$$

For $m \neq 0$, we seek trial solutions of the form $R(r) = r^k$. Substituting this into the radial equation gives

$$r^2k(k-1)r^{k-2} + rkr^{k-1} - m^2r^k = 0$$

Simplifying the exponents yields

$$k(k-1)r^k + kr^k - m^2 r^k = 0$$

Dividing by r^k gives the characteristic equation

$$k(k-1) + k - m^2 = 0$$

Expanding and canceling terms gives

$$k^2 - m^2 = 0$$

The roots of this algebraic equation are $k = \pm m$. Thus, for $m > 0$, the general solution for the radial function is

$$R_m(r) = C_m r^m + D_m r^{-m}$$

For the case $m = 0$, the radial equation simplifies to

$$r^2 \frac{d^2 R_0}{dr^2} + r \frac{dR_0}{dr} = 0$$

Dividing by r gives

$$r \frac{d^2 R_0}{dr^2} + \frac{dR_0}{dr} = 0$$

Rewriting the left-hand side as a total derivative yields

$$\frac{d}{dr}\left(r \frac{dR_0}{dr} \right) = 0$$

Integrating with respect to r gives

$$r \frac{dR_0}{dr} = D_0$$

Dividing by r yields

$$\frac{dR_0}{dr} = \frac{D_0}{r}$$

Integrating a second time gives the radial solution for $m = 0$

$$R_0(r) = C_0 + D_0 \ln r$$

The general solution for the potential in cylindrical symmetry is obtained by superposing all allowed modes m

$$V(r, \phi) = C_0 + D_0 \ln r + \sum_{m=1}^{\infty} \left(C_m r^m + D_m r^{-m} \right) \left[A_m \cos(m\phi) + B_m \sin(m\phi) \right]$$

Laplace's Equation in Spherical Coordinates

For systems possessing spherical symmetry with azimuthal independence (where $\frac{\partial V}{\partial \phi} = 0$), Laplace's equation in spherical coordinates (r, θ, ϕ) reduces to

$$\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial V}{\partial r}\right)+\frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial V}{\partial\theta}\right)=0$$

We separate variables by writing the potential as the product of a radial function and an angular function

$$V(r,\theta)=R(r)\Theta(\theta)$$

Substituting this product into Laplace's equation gives

$$\frac{\Theta(\theta)}{r^2}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right)+\frac{R(r)}{r^2\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right)=0$$

Multiplying the equation by $\frac{r^2}{R(r)\Theta(\theta)}$ isolates the terms

$$\frac{1}{R(r)}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right)+\frac{1}{\Theta(\theta)\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right)=0$$

Moving the angular terms to the right-hand side separates the variables

$$\frac{1}{R(r)}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right)=-\frac{1}{\Theta(\theta)\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right)$$

We set both sides equal to a dimensionless separation constant, conventionally written as $l(l+1)$. Equating the radial part to $l(l+1)$ gives

$$\frac{1}{R(r)}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right)=l(l+1)$$

Multiplying by $R(r)$ yields

$$\frac{d}{dr}\left(r^2\frac{dR}{dr}\right)-l(l+1)R(r)=0$$

Carrying out the derivative on the first term gives

$$r^2\frac{d^2R}{dr^2}+2r\frac{dR}{dr}-l(l+1)R(r)=0$$

We substitute a power solution of the form $R(r)=r^k$ into this radial equation

$$r^2k(k-1)r^{k-2}+2rkr^{k-1}-l(l+1)r^k=0$$

Simplifying the powers of r yields

$$k(k-1)r^k+2kr^k-l(l+1)r^k=0$$

Dividing by r^k gives the algebraic characteristic equation

$$k(k-1)+2k-l(l+1)=0$$

Expanding the brackets yields

$$k^2+k-l(l+1)=0$$

Factoring the quadratic expression in k gives

$$(k-l)(k+l+1)=0$$

The two roots of this characteristic equation are $k=l$ and $k=-(l+1)$. Therefore, the radial part gives linear combinations of integer powers of r

$$R_l(r)=A_l r^l+B_l r^{-(l+1)}$$

Equating the angular part of Laplace's equation to $l(l+1)$ gives

$$-\frac{1}{\Theta(\theta)\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right)=l(l+1)$$

Multiplying by $-\Theta(\theta)$ yields

$$\frac{1}{\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right)+l(l+1)\Theta(\theta)=0$$

To transform this differential equation into standard form, we define a change of variable $x=\cos\theta$. The differential operators transform according to $\frac{d}{d\theta}=-\sin\theta\frac{d}{dx}$, and using $\sin^2\theta=1-x^2$, the equation becomes

$$\frac{d}{dx}\left((1-x^2)\frac{d\Theta}{dx}\right)+l(l+1)\Theta(x)=0$$

Carrying out the derivative yields Legendre's differential equation

$$(1-x^2)\frac{d^2\Theta}{dx^2}-2x\frac{d\Theta}{dx}+l(l+1)\Theta(x)=0$$

For the solution to remain finite and single-valued across the physical domain $\theta\in[0,\pi]$, which corresponds to $x\in[-1,1]$, the parameter l must be a non-negative integer $l=0,1,2,\ldots$. Under this condition, the acceptable angular solutions are the Legendre polynomials $\Theta_l(\theta)=P_l(\cos\theta)$. The complete general solution for azimuthally symmetric potentials in spherical coordinates is therefore written as

$$V(r,\theta)=\sum_{l=0}^{\infty}\left(A_l r^l+B_l r^{-(l+1)}\right)P_l(\cos\theta)$$

4. The Wave Equation and Vibrational Modes of a Stretched String

Derivation of the Wave Equation from Newton's Second Law

Consider a uniform, perfectly flexible string of linear mass density μ stretched under a constant tension T along the x-axis. We examine small transverse displacements $y(x,t)$ of the string from its equilibrium position.

Let us isolate a small segment of the string located between x and $x+\Delta x$. The tension T acts tangentially along the string at both endpoints of the segment. At position $x+\Delta x$, the tension vector makes an angle $\theta(x+\Delta x,t)$ with the horizontal, contributing an upward transverse force component

$$F_{y,\text{right}}=T\sin\theta(x+\Delta x,t)$$

At position x, the tension vector acts at an angle $\theta(x,t)$ with the horizontal, contributing a downward transverse force component

$$F_{y,\text{left}} = -T\sin\theta(x,t)$$

The net transverse force F_y acting on the small element is the sum of these two contributions

$$F_y = T\sin\theta(x+\Delta x,t) - T\sin\theta(x,t)$$

For small transverse vibrations, the angle θ is very small, allowing us to use the small-angle approximation $\sin\theta \approx \tan\theta$. Since the slope of the string at any point is given by the spatial partial derivative $\frac{\partial y}{\partial x}$, we have

$$\sin\theta \approx \frac{\partial y}{\partial x}$$

Substituting this slope approximation into the net force expression gives

$$F_y = T\left[\left.\frac{\partial y}{\partial x}\right|_{x+\Delta x} - \left.\frac{\partial y}{\partial x}\right|_{x}\right]$$

The mass Δm of the string element of length Δx is given by

$$\Delta m = \mu \Delta x$$

According to Newton's second law of motion, the net transverse force on the element equals its mass multiplied by its transverse acceleration $\frac{\partial^2 y}{\partial t^2}$

$$T\left[\left.\frac{\partial y}{\partial x}\right|_{x+\Delta x} - \left.\frac{\partial y}{\partial x}\right|_{x}\right] = \mu \Delta x \frac{\partial^2 y}{\partial t^2}$$

Dividing both sides of this equation by Δx yields

$$T\frac{\left.\frac{\partial y}{\partial x}\right|_{x+\Delta x} - \left.\frac{\partial y}{\partial x}\right|_{x}}{\Delta x} = \mu\frac{\partial^2 y}{\partial t^2}$$

Taking the limit as $\Delta x \to 0$, the left-hand side reduces by definition to the second spatial derivative of $y(x,t)$

$$T\frac{\partial^2 y}{\partial x^2} = \mu\frac{\partial^2 y}{\partial t^2}$$

Dividing both sides by T isolates the spatial derivative

$$\frac{\partial^2 y}{\partial x^2} = \frac{\mu}{T}\frac{\partial^2 y}{\partial t^2}$$

Defining the phase velocity of wave propagation along the string as $v = \sqrt{\frac{T}{\mu}}$, we obtain $\frac{1}{v^2} = \frac{\mu}{T}$. Substituting this into the differential equation gives the standard one-dimensional wave equation

$$\frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 y}{\partial t^2}$$

Solution by Separation of Variables

To solve the one-dimensional wave equation for a string of length L fixed at both ends, we assume a separable solution of the form

$$y(x,t) = X(x)G(t)$$

Substituting this product into the wave equation gives

$$\frac{d^2X}{dx^2}G(t) = \frac{1}{v^2}X(x)\frac{d^2G}{dt^2}$$

Dividing both sides by $X(x)G(t)$ separates the spatial dependence from the temporal dependence

$$\frac{1}{X(x)}\frac{d^2X}{dx^2} = \frac{1}{v^2G(t)}\frac{d^2G}{dt^2}$$

Because the left side depends only on x and the right side depends only on t, both sides must equal a negative separation constant, which we denote as $-k^2$. Setting the spatial part to $-k^2$ gives

$$\frac{1}{X(x)}\frac{d^2X}{dx^2} = -k^2$$

Multiplying by $X(x)$ yields the spatial ordinary differential equation

$$\frac{d^2X}{dx^2} + k^2X(x) = 0$$

Setting the temporal part to $-k^2$ gives

$$\frac{1}{v^2G(t)}\frac{d^2G}{dt^2} = -k^2$$

Multiplying by $v^2G(t)$ yields the temporal ordinary differential equation

$$\frac{d^2G}{dt^2} + k^2v^2G(t) = 0$$

Defining the angular frequency $\omega = kv$, the temporal differential equation becomes

$$\frac{d^2G}{dt^2} + \omega^2G(t) = 0$$

Boundary Conditions, Normal Modes, and Frequencies of Vibration

The general solution for the spatial differential equation is

$$X(x) = A\cos(kx) + B\sin(kx)$$

For a string fixed at both boundaries $x = 0$ and $x = L$, the displacement must vanish at these points for all time t, imposing the boundary conditions $X(0) = 0$ and $X(L) = 0$. Applying the first boundary condition at $x = 0$ yields

$$X(0) = A\cos(0) + B\sin(0) = 0$$

Since $\cos(0) = 1$ and $\sin(0) = 0$, this requires

$$A = 0$$

The spatial function reduces to

$$X(x) = B\sin(kx)$$

Applying the second boundary condition at $x = L$ yields

$$X(L) = B\sin(kL) = 0$$

To obtain non-trivial solutions ($B \neq 0$), we must have

$$\sin(kL) = 0$$

This condition restricts the allowable wave numbers k to discrete quantized values k_n given by

$$k_n L = n\pi \implies k_n = \frac{n\pi}{L}, \quad n = 1, 2, 3, \ldots$$

The spatial profile corresponding to the n-th standing wave mode, known as a normal mode, is given by

$$X_n(x) = \sin\left(\frac{n\pi x}{L}\right)$$

The general solution to the temporal equation for the n-th mode is

$$G_n(t) = C_n\cos(\omega_n t) + D_n\sin(\omega_n t)$$

where the discrete angular frequencies ω_n are given by

$$\omega_n = k_n v = \frac{n\pi v}{L}$$

The corresponding natural frequencies of vibration $f_n = \frac{\omega_n}{2\pi}$ for the n-th normal mode are

$$f_n = \frac{nv}{2L}$$

Substituting $v = \sqrt{\frac{T}{\mu}}$ gives Mersenne's law for the vibrational frequencies of a stretched string

$$f_n = \frac{n}{2L}\sqrt{\frac{T}{\mu}}, \quad n = 1, 2, 3, \ldots$$

The fundamental frequency (or first harmonic) occurs for $n = 1$

$$f_1 = \frac{1}{2L}\sqrt{\frac{T}{\mu}}$$

Higher modes ($n = 2, 3, \ldots$) represent higher harmonics whose frequencies are integer multiples of the fundamental frequency, $f_n = nf_1$.

General Solution as a Sum of Modes

By the principle of linear superposition, the general solution to the wave equation is an arbitrary sum over all possible normal modes n

$$y(x,t) = \sum_{n=1}^{\infty} X_n(x) G_n(t)$$

Substituting the spatial and temporal functions explicitly gives the general solution

$$y(x,t) = \sum_{n=1}^{\infty} \sin\left(\frac{n\pi x}{L}\right) \left[C_n \cos\left(\frac{n\pi v t}{L}\right) + D_n \sin\left(\frac{n\pi v t}{L}\right) \right]$$

The constant coefficients C_n and D_n are determined completely by the initial displacement profile $y(x,0) = f(x)$ and initial velocity profile $\left.\frac{\partial y}{\partial t}\right|_{t=0} = g(x)$ using the orthogonality properties of the sine functions.

5. Vibrational Modes of Rectangular and Circular Membranes

The transverse vibrations of a thin, flexible membrane clamped at its boundaries are governed by the two-dimensional wave equation. We first examine a membrane with a rectangular boundary and then extend the separation of variables method to a membrane with a circular boundary.

Rectangular Membranes

Consider a thin membrane stretched over a rectangular region in the xy-plane with sides of length a along the x-axis and b along the y-axis. The membrane is fixed at all four edges. The transverse displacement $u(x,y,t)$ at position (x,y) and time t satisfies the two-dimensional wave equation:

$$\frac{\partial^2 u}{\partial t^2} = v^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right)$$

where $v = \sqrt{T/\sigma}$ is the speed of transverse waves in the membrane, T is the uniform surface tension, and σ is the mass per unit area.

The boundary conditions representing fixed edges are:

$$u(0,y,t) = 0$$

$$u(a,y,t) = 0$$

$$u(x,0,t) = 0$$

$$u(x,b,t) = 0$$

To solve the wave equation, we assume a separable solution of the form:

$$u(x,y,t) = X(x)Y(y)T(t)$$

Substituting this product form into the two-dimensional wave equation yields:

$$X(x)Y(y)\frac{d^2T}{dt^2} = v^2\left(Y(y)T(t)\frac{d^2X}{dx^2} + X(x)T(t)\frac{d^2Y}{dy^2}\right)$$

Dividing both sides of this equation by $v^2X(x)Y(y)T(t)$ separates the spatial variables from the temporal variable:

$$\frac{1}{v^2T(t)}\frac{d^2T}{dt^2} = \frac{1}{X(x)}\frac{d^2X}{dx^2} + \frac{1}{Y(y)}\frac{d^2Y}{dy^2}$$

Since the left side depends only on t and the right side depends only on x and y, both sides must equal a negative real separation constant, which we denote as $-\lambda^2$. This yields the ordinary differential equation for time:

$$\frac{d^2T}{dt^2} + v^2\lambda^2T = 0$$

Defining the angular frequency as $\omega = v\lambda$, the time equation becomes:

$$\frac{d^2T}{dt^2} + \omega^2T = 0$$

The remaining spatial part is given by:

$$\frac{1}{X(x)}\frac{d^2X}{dx^2} + \frac{1}{Y(y)}\frac{d^2Y}{dy^2} = -\lambda^2$$

Rearranging this equation isolates the x-dependence on one side:

$$\frac{1}{X(x)}\frac{d^2X}{dx^2} = -\lambda^2 - \frac{1}{Y(y)}\frac{d^2Y}{dy^2}$$

Because the left side is a function of x alone and the right side is a function of y alone, each side must equal a constant, denoted by $-k_x^2$. This gives two spatial ordinary differential equations:

$$\frac{d^2X}{dx^2} + k_x^2X = 0$$

$$\frac{d^2Y}{dy^2} + k_y^2Y = 0$$

where the separation constants satisfy the relation:

$$k_x^2 + k_y^2 = \lambda^2$$

The general solution for $X(x)$ is:

$$X(x) = A_1\sin(k_xx) + B_1\cos(k_xx)$$

Applying the boundary condition at $x = 0$ requires $X(0) = 0$, which yields:

$$B_1 = 0$$

Applying the boundary condition at $x = a$ requires $X(a) = 0$, leading to:

$$\sin(k_x a) = 0$$

Thus, the allowed wave numbers in the x-direction are constrained to non-zero integer multiples of π/a:

$$k_x = \frac{m\pi}{a}, \quad m = 1, 2, 3, \ldots$$

Similarly, the general solution for $Y(y)$ is:

$$Y(y) = A_2 \sin(k_y y) + B_2 \cos(k_y y)$$

Applying the boundary condition at $y = 0$ gives $Y(0) = 0$, which requires:

$$B_2 = 0$$

Applying the boundary condition at $y = b$ gives $Y(b) = 0$, leading to:

$$\sin(k_y b) = 0$$

Thus, the allowed wave numbers in the y-direction are constrained to non-zero integer multiples of π/b:

$$k_y = \frac{n\pi}{b}, \quad n = 1, 2, 3, \ldots$$

Substituting the expressions for k_x and k_y into the relation $\lambda_{mn}^2 = k_x^2 + k_y^2$ gives:

$$\lambda_{mn} = \sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2}$$

The corresponding angular frequencies $\omega_{mn} = v\lambda_{mn}$ of the normal modes are:

$$\omega_{mn} = v\pi\sqrt{\frac{m^2}{a^2} + \frac{n^2}{b^2}}$$

The mode frequencies $f_{mn} = \omega_{mn}/(2\pi)$ of the rectangular membrane are given by:

$$f_{mn} = \frac{v}{2}\sqrt{\frac{m^2}{a^2} + \frac{n^2}{b^2}}$$

The spatial profile of the normal mode specified by the mode numbers (m, n) is:

$$u_{mn}(x, y, t) = \sin\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) \left[C_{mn}\cos(\omega_{mn} t) + D_{mn}\sin(\omega_{mn} t)\right]$$

Circular Membranes

For a circular membrane of radius a clamped at its outer perimeter, it is natural to use plane polar coordinates (r, θ). The relation between Cartesian and polar coordinates is $x = r\cos\theta$ and $y = r\sin\theta$. In polar coordinates, the two-dimensional wave equation for the transverse displacement $u(r, \theta, t)$ is expressed as:

$$\frac{\partial^2 u}{\partial t^2} = v^2\left(\frac{\partial^2 u}{\partial r^2} + \frac{1}{r}\frac{\partial u}{\partial r} + \frac{1}{r^2}\frac{\partial^2 u}{\partial \theta^2}\right)$$

The fixed boundary condition at the outer edge is:

$$u(a, \theta, t) = 0$$

Additionally, the physical solution must remain finite at the center $r = 0$, and must be periodic in the angular coordinate with period 2π:

$$u(r, \theta + 2\pi, t) = u(r, \theta, t)$$

We seek solutions by separation of variables using the form:

$$u(r, \theta, t) = R(r)\Theta(\theta)T(t)$$

Substituting this product solution into the polar wave equation yields:

$$R(r)\Theta(\theta)\frac{d^2T}{dt^2} = v^2\left(\Theta(\theta)T(t)\frac{d^2R}{dr^2} + \frac{\Theta(\theta)T(t)}{r}\frac{dR}{dr} + \frac{R(r)T(t)}{r^2}\frac{d^2\Theta}{d\theta^2}\right)$$

Dividing both sides by $v^2R(r)\Theta(\theta)T(t)$ isolates the temporal dependence:

$$\frac{1}{v^2T(t)}\frac{d^2T}{dt^2} = \frac{1}{R(r)}\frac{d^2R}{dr^2} + \frac{1}{rR(r)}\frac{dR}{dr} + \frac{1}{r^2\Theta(\theta)}\frac{d^2\Theta}{d\theta^2}$$

Setting both sides equal to $-k^2$ gives the temporal differential equation:

$$\frac{d^2T}{dt^2} + \omega^2 T = 0$$

where $\omega = vk$.

The remaining spatial differential equation becomes:

$$\frac{1}{R(r)}\frac{d^2R}{dr^2} + \frac{1}{rR(r)}\frac{dR}{dr} + \frac{1}{r^2\Theta(\theta)}\frac{d^2\Theta}{d\theta^2} = -k^2$$

Multiplying this entire equation by r^2 yields:

$$\frac{r^2}{R(r)}\frac{d^2R}{dr^2} + \frac{r}{R(r)}\frac{dR}{dr} + k^2r^2 = -\frac{1}{\Theta(\theta)}\frac{d^2\Theta}{d\theta^2}$$

Since the left side depends only on r and the right side depends only on θ, both sides must equal a constant, which we denote as m^2. The angular equation is:

$$\frac{d^2\Theta}{d\theta^2} + m^2\Theta = 0$$

The general solution for the angular dependence is:

$$\Theta(\theta) = A_m\cos(m\theta) + B_m\sin(m\theta)$$

The requirement that $\Theta(\theta + 2\pi) = \Theta(\theta)$ restricts m to non-negative integers:

$$m = 0, 1, 2, 3, \ldots$$

Setting the radial side equal to m^2 yields:

$$\frac{r^2}{R(r)}\frac{d^2R}{dr^2} + \frac{r}{R(r)}\frac{dR}{dr} + k^2r^2 = m^2$$

Multiplying by $R(r)$ gives the radial differential equation:

$$r^2\frac{d^2R}{dr^2} + r\frac{dR}{dr} + (k^2r^2 - m^2)R = 0$$

To transform this into standard form, we define a dimensionless variable $s = kr$. By the chain rule, the spatial derivatives transform as:

$$\frac{dR}{dr} = \frac{dR}{ds}\frac{ds}{dr} = k\frac{dR}{ds}$$
$$\frac{d^2R}{dr^2} = \frac{d}{dr}\left(k\frac{dR}{ds}\right) = k^2\frac{d^2R}{ds^2}$$

Substituting these derivative expressions into the radial differential equation gives:

$$r^2\left(k^2\frac{d^2R}{ds^2}\right) + r\left(k\frac{dR}{ds}\right) + (k^2r^2 - m^2)R = 0$$

Using $s = kr$, this equation simplifies directly to Bessel's differential equation of order m:

$$s^2\frac{d^2R}{ds^2} + s\frac{dR}{ds} + (s^2 - m^2)R = 0$$

This equation has the general solution:

$$R(s) = C_1J_m(s) + C_2Y_m(s)$$

where $J_m(s)$ is the Bessel function of the first kind of order m, and $Y_m(s)$ is the Bessel function of the second kind (Neumann function) of order m.

Because the displacement at the center $r = 0$ must remain finite, and $Y_m(s) \to -\infty$ as $s \to 0$, we must set $C_2 = 0$. Hence, the radial function is given by:

$$R(r) = C_1J_m(kr)$$

Applying the boundary condition at the outer rim $r = a$ requires $R(a) = 0$:

$$J_m(ka) = 0$$

Let α_{mn} denote the n-th positive zero of the Bessel function $J_m(x)$, where $n = 1, 2, 3, \ldots$. The allowed values for k are therefore quantized:

$$k_{mn} = \frac{\alpha_{mn}}{a}$$

The corresponding vibrational frequencies of the normal modes of the circular membrane are given by:

$$\omega_{mn} = vk_{mn} = \frac{v\alpha_{mn}}{a}$$

The mode frequencies $f_{mn} = \omega_{mn}/(2\pi)$ are:

$$f_{mn} = \frac{v\alpha_{mn}}{2\pi a}$$

The complete modal solution for the displacement of a circular membrane corresponding to indices (m, n) is:

$$u_{mn}(r, \theta, t) = J_m\left(\frac{\alpha_{mn} r}{a}\right)\left[A_{mn}\cos(m\theta) + B_{mn}\sin(m\theta)\right]\left[C_{mn}\cos(\omega_{mn}t) + D_{mn}\sin(\omega_{mn}t)\right]$$

For radially symmetric modes ($m = 0$), the solution has no angular dependence, and the nodal lines consist purely of concentric circles corresponding to the zeros of $J_0(\alpha_{0n} r/a)$. For $m > 0$, the nodal patterns consist of both concentric circles and radial lines.

6. The Diffusion Equation

The diffusion equation describes transport processes driven by concentration gradients or thermal conduction without macroscopically ordered fluid motion. In one spatial dimension, the equation for a field $u(x, t)$ (such as temperature or particle concentration) is given by:

$$\frac{\partial u}{\partial t} = D\frac{\partial^2 u}{\partial x^2}$$

where $D > 0$ is the diffusion constant (or thermal diffusivity).

Solution by Separation of Variables

Consider a uniform rod of length L along the x-axis from $x = 0$ to $x = L$, with its ends maintained at a constant zero boundary condition:

$$u(0, t) = 0$$

$$u(L, t) = 0$$

We seek a separable solution of the form:

$$u(x, t) = X(x)T(t)$$

Substituting this product into the one-dimensional diffusion equation yields:

$$X(x)\frac{dT}{dt} = DT(t)\frac{d^2X}{dx^2}$$

Dividing both sides by $DX(x)T(t)$ separates the variables:

$$\frac{1}{DT(t)}\frac{dT}{dt} = \frac{1}{X(x)}\frac{d^2X}{dx^2}$$

Since the left side depends only on t and the right side depends only on x, both sides must equal a common constant, which we set to $-\lambda^2$ to ensure physically meaningful solutions that do not grow unbounded. This gives two separate ordinary differential equations:

$$\frac{d^2X}{dx^2} + \lambda^2 X = 0$$

$$\frac{dT}{dt} + D\lambda^2 T = 0$$

The general solution of the spatial equation is:

$$X(x) = A\sin(\lambda x) + B\cos(\lambda x)$$

Applying the boundary condition at $x = 0$ requires $X(0) = 0$:

$$B = 0$$

Applying the boundary condition at $x = L$ requires $X(L) = 0$:

$$A\sin(\lambda L) = 0$$

To obtain a non-trivial solution ($A \neq 0$), we must have:

$$\sin(\lambda L) = 0$$

This determines the allowed spatial eigenvalues λ_n:

$$\lambda_n = \frac{n\pi}{L}, \quad n = 1, 2, 3, \ldots$$

The spatial eigenfunctions are thus:

$$X_n(x) = \sin\left(\frac{n\pi x}{L}\right)$$

Now substituting λ_n into the temporal differential equation gives:

$$\frac{dT}{dt} = -D\left(\frac{n\pi}{L}\right)^2 T$$

Integrating this first-order ordinary differential equation with respect to t yields:

$$\int \frac{1}{T} dT = -D\left(\frac{n\pi}{L}\right)^2 \int dt$$

$$\ln T(t) = -D\left(\frac{n\pi}{L}\right)^2 t + \text{constant}$$

Exponentiating both sides gives the temporal solution for the n-th mode:

$$T_n(t) = C_n \exp\left[-D\left(\frac{n\pi}{L}\right)^2 t\right]$$

Thus, the fundamental product solutions (modes) of the diffusion equation are:

$$u_n(x,t) = A_n \sin\left(\frac{n\pi x}{L}\right) \exp\left[-D\left(\frac{n\pi}{L}\right)^2 t\right]$$

Time Dependence: Decay vs. Oscillation

The time dependence of the diffusion equation differs fundamentally from that of the wave equation due to the order of the temporal derivative in the governing partial differential equation.

The one-dimensional wave equation features a second-order derivative in time:

$$\frac{\partial^2 u}{\partial t^2} = v^2 \frac{\partial^2 u}{\partial x^2}$$

Separation of variables for the wave equation leads to a second-order temporal ordinary differential equation:

$$\frac{d^2T}{dt^2} + \omega^2 T = 0$$

The general solution to this second-order harmonic equation is linear combinations of oscillatory functions:

$$T(t) = C_1 \cos(\omega t) + C_2 \sin(\omega t) = Ce^{i\omega t} + De^{-i\omega t}$$

This describes persistent, reversible oscillatory motion where energy exchanges continuously between kinetic and potential forms without dissipation.

In contrast, the diffusion equation features a first-order derivative in time:

$$\frac{\partial u}{\partial t} = D \frac{\partial^2 u}{\partial x^2}$$

Separation of variables leads to a first-order temporal ordinary differential equation:

$$\frac{dT}{dt} + D\lambda^2 T = 0$$

The solution to this equation is a purely decaying real exponential:

$$T(t) = T(0)e^{-\Gamma t}$$

where the decay rate is given by $\Gamma = D\lambda^2 = D(n\pi/L)^2$.

Because the exponent $-\Gamma t$ is real and negative, the amplitude of every spatial Fourier component decays monotonically to zero as $t \to \infty$. Higher spatial modes (larger n) decay more rapidly due to the n^2 factor in the decay rate Γ. Consequently, the diffusion equation models irreversible, dissipative relaxation processes toward thermal or concentration equilibrium, without any non-decaying oscillation.

7. Hermite Polynomials and Recurrence Relations

Hermite polynomials $H_n(x)$ are a set of classical orthogonal polynomials defined over the interval $(-\infty, \infty)$ with respect to the weight function $w(x) = e^{-x^2}$. They can be explicitly defined using Rodrigues' formula:

$$H_n(x) = (-1)^n e^{x^2} \frac{d^n}{dx^n}\left(e^{-x^2}\right)$$

First Few Hermite Polynomials

The first few Hermite polynomials are evaluated directly from Rodrigues' formula for low values of n.

For $n = 0$, substituting $n = 0$ into Rodrigues' formula gives:

$$H_0(x) = (-1)^0 e^{x^2} \frac{d^0}{dx^0}\left(e^{-x^2}\right) = 1 \cdot e^{x^2} \cdot e^{-x^2} = 1$$

For $n = 1$, taking the first derivative of e^{-x^2} with respect to x yields:

$$\frac{d}{dx}\left(e^{-x^2}\right) = -2xe^{-x^2}$$

Substituting this derivative into Rodrigues' formula for $n = 1$ gives:

$$H_1(x) = (-1)^1 e^{x^2}\left(-2xe^{-x^2}\right) = 2x$$

For $n = 2$, taking the derivative of $-2xe^{-x^2}$ with respect to x gives:

$$\frac{d^2}{dx^2}\left(e^{-x^2}\right) = \frac{d}{dx}\left(-2xe^{-x^2}\right) = -2e^{-x^2} + 4x^2e^{-x^2}$$

Substituting this second derivative into Rodrigues' formula for $n = 2$ gives:

$$H_2(x) = (-1)^2 e^{x^2}\left(4x^2 - 2\right)e^{-x^2} = 4x^2 - 2$$

For $n = 3$, taking the derivative of $\left(4x^2 - 2\right)e^{-x^2}$ with respect to x gives:

$$\frac{d^3}{dx^3}\left(e^{-x^2}\right) = \frac{d}{dx}\left[\left(4x^2 - 2\right)e^{-x^2}\right] = 8xe^{-x^2} - 2x\left(4x^2 - 2\right)e^{-x^2} = \left(12x - 8x^3\right)e^{-x^2}$$

Substituting this third derivative into Rodrigues' formula for $n = 3$ gives:

$$H_3(x) = (-1)^3 e^{x^2}\left(12x - 8x^3\right)e^{-x^2} = 8x^3 - 12x$$

Recurrence Relations of Hermite Polynomials

We derive the primary recurrence relations of Hermite polynomials directly using Rodrigues' formula and Leibniz's rule.

Let $u = e^{-x^2}$. Differentiating u once with respect to x gives:

$$\frac{du}{dx} = -2xe^{-x^2} = -2xu$$

Rearranging this derivative yields the linear differential equation:

$$\frac{du}{dx} + 2xu = 0$$

Differentiating this differential equation n times with respect to x using Leibniz's rule for the n-th derivative of a product gives:

$$\frac{d^{n+1}u}{dx^{n+1}} + 2\left(x\frac{d^n u}{dx^n} + n\frac{d^{n-1}u}{dx^{n-1}}\right) = 0$$

Multiplying the entire equation by $(-1)^n e^{x^2}$ yields:

$$(-1)^n e^{x^2}\frac{d^{n+1}u}{dx^{n+1}} + 2x(-1)^n e^{x^2}\frac{d^n u}{dx^n} + 2n(-1)^n e^{x^2}\frac{d^{n-1}u}{dx^{n-1}} = 0$$

We re-express each term in terms of Hermite polynomials using the Rodrigues identity $H_k(x) = (-1)^k e^{x^2}\frac{d^k u}{dx^k}$.

The first term is rewritten by factoring out a minus sign as $(-1)^n = -(-1)^{n+1}$:

$$(-1)^n e^{x^2}\frac{d^{n+1}u}{dx^{n+1}} = -(-1)^{n+1}e^{x^2}\frac{d^{n+1}u}{dx^{n+1}} = -H_{n+1}(x)$$

The second term directly matches the definition of $H_n(x)$:

$$2x(-1)^n e^{x^2}\frac{d^n u}{dx^n} = 2xH_n(x)$$

The third term is rewritten by factoring out a minus sign as $(-1)^n = -(-1)^{n-1}$:

$$2n(-1)^n e^{x^2}\frac{d^{n-1}u}{dx^{n-1}} = -2n(-1)^{n-1}e^{x^2}\frac{d^{n-1}u}{dx^{n-1}} = -2nH_{n-1}(x)$$

Substituting these three expressions back into the differentiated equation yields:

$$-H_{n+1}(x) + 2xH_n(x) - 2nH_{n-1}(x) = 0$$

Rearranging the terms produces the three-term recurrence relation:

$$2xH_n(x) = 2nH_{n-1}(x) + H_{n+1}(x)$$

Next, we establish the derivative recurrence relation $H_n'(x) = 2nH_{n-1}(x)$.

Differentiating Rodrigues' formula $H_n(x) = (-1)^n e^{x^2}\frac{d^n u}{dx^n}$ with respect to x using the product rule gives:

$$H_n'(x) = (-1)^n\left[2xe^{x^2}\frac{d^n u}{dx^n} + e^{x^2}\frac{d^{n+1}u}{dx^{n+1}}\right]$$

Distributing $(-1)^n$ into both terms yields:

$$H_n'(x) = 2x\left[(-1)^n e^{x^2}\frac{d^n u}{dx^n}\right] - \left[(-1)^{n+1}e^{x^2}\frac{d^{n+1}u}{dx^{n+1}}\right]$$

Substituting the definitions of $H_n(x)$ and $H_{n+1}(x)$ into the right-hand side gives:

$$H_n'(x) = 2xH_n(x) - H_{n+1}(x)$$

From the three-term recurrence relation $2xH_n(x) = 2nH_{n-1}(x) + H_{n+1}(x)$, we isolate $H_{n+1}(x)$ as:

$$H_{n+1}(x) = 2xH_n(x) - 2nH_{n-1}(x)$$

Substituting this expression for $H_{n+1}(x)$ into the derivative relation yields:

$$H_n'(x) = 2xH_n(x) - [2xH_n(x) - 2nH_{n-1}(x)]$$

Simplifying the right-hand side yields the derivative recurrence relation:

$$H_n'(x) = 2nH_{n-1}(x)$$

8. Generating Function of Hermite Polynomials

The generating function $g(x,t)$ for Hermite polynomials is given by e^{2xt-t^2}. We prove that:

$$e^{2xt-t^2} = \sum_{n=0}^{\infty} H_n(x)\frac{t^n}{n!}$$

Consider the exponential function $f(t) = e^{-(x-t)^2}$ as a function of t.

Expanding $f(t)$ as a Taylor series in powers of t about $t = 0$ gives:

$$f(t) = \sum_{n=0}^{\infty} \left[\frac{d^n}{dt^n}\left(e^{-(x-t)^2}\right)\right]_{t=0} \frac{t^n}{n!}$$

We define a change of variable $y = x - t$.

Differentiating y with respect to t gives $\frac{dy}{dt} = -1$, which implies the operator identity:

$$\frac{d}{dt} = -\frac{d}{dy}$$

Applying this operator n times to e^{-y^2} gives:

$$\frac{d^n}{dt^n}\left(e^{-(x-t)^2}\right) = (-1)^n \frac{d^n}{dy^n}\left(e^{-y^2}\right)$$

Evaluating this n-th derivative at $t = 0$ corresponds to setting $y = x$:

$$\left[\frac{d^n}{dt^n}\left(e^{-(x-t)^2}\right)\right]_{t=0} = (-1)^n \frac{d^n}{dx^n}\left(e^{-x^2}\right)$$

Now, we rewrite the exponent of the generating function e^{2xt-t^2} by completing the square in t:

$$2xt - t^2 = x^2 - (x^2 - 2xt + t^2) = x^2 - (x-t)^2$$

Exponentiating both sides of this algebraic identity yields:

$$e^{2xt-t^2} = e^{x^2-(x-t)^2} = e^{x^2}e^{-(x-t)^2}$$

Substituting the Taylor series expansion of $e^{-(x-t)^2}$ into this product yields:

$$e^{2xt-t^2} = e^{x^2}\sum_{n=0}^{\infty} \left[\frac{d^n}{dt^n}\left(e^{-(x-t)^2}\right)\right]_{t=0} \frac{t^n}{n!}$$

Substituting the evaluated derivative at $t = 0$ into the series gives:

$$e^{2xt-t^2} = e^{x^2} \sum_{n=0}^{\infty} \left((-1)^n \frac{d^n}{dx^n} \left(e^{-x^2} \right) \right) \frac{t^n}{n!}$$

Bringing the factor e^{x^2} inside the summation yields:

$$e^{2xt-t^2} = \sum_{n=0}^{\infty} \left((-1)^n e^{x^2} \frac{d^n}{dx^n} \left(e^{-x^2} \right) \right) \frac{t^n}{n!}$$

Recognizing the term inside the parentheses as Rodrigues' formula $H_n(x) = (-1)^n e^{x^2} \frac{d^n}{dx^n} \left(e^{-x^2} \right)$ completes the proof:

$$e^{2xt-t^2} = \sum_{n=0}^{\infty} H_n(x) \frac{t^n}{n!}$$

9. Laguerre Polynomials and Recurrence Relations

Laguerre polynomials $L_n(x)$ are solutions to Laguerre's differential equation and are orthogonal over the interval $[0, \infty)$ with weight function $w(x) = e^{-x}$. They are defined by Rodrigues' formula:

$$L_n(x) = \frac{e^x}{n!} \frac{d^n}{dx^n} \left(x^n e^{-x} \right)$$

First Few Laguerre Polynomials

We evaluate the first few Laguerre polynomials using Rodrigues' formula for low values of n.

For $n = 0$, substituting $n = 0$ into Rodrigues' formula gives:

$$L_0(x) = \frac{e^x}{0!} \frac{d^0}{dx^0} \left(x^0 e^{-x} \right) = e^x \cdot e^{-x} = 1$$

For $n = 1$, taking the first derivative of xe^{-x} with respect to x using the product rule gives:

$$\frac{d}{dx} \left(xe^{-x} \right) = e^{-x} - xe^{-x} = (1 - x)e^{-x}$$

Substituting this derivative into Rodrigues' formula for $n = 1$ yields:

$$L_1(x) = \frac{e^x}{1!}(1 - x)e^{-x} = 1 - x$$

For $n = 2$, taking the first derivative of x^2e^{-x} gives:

$$\frac{d}{dx} \left(x^2e^{-x} \right) = 2xe^{-x} - x^2e^{-x}$$

Taking the second derivative of x^2e^{-x} gives:

$$\frac{d^2}{dx^2} \left(x^2e^{-x} \right) = \frac{d}{dx} \left(2xe^{-x} - x^2e^{-x} \right) = 2e^{-x} - 2xe^{-x} - \left(2xe^{-x} - x^2e^{-x} \right) = \left(2 - 4x + x^2 \right) e^{-x}$$

Substituting this second derivative into Rodrigues' formula for $n = 2$ gives:

$$L_2(x) = \frac{e^x}{2!}\left(2 - 4x + x^2\right)e^{-x} = \frac{1}{2}\left(2 - 4x + x^2\right) = 1 - x + \frac{x^2}{2}$$

Recurrence Relations of Laguerre Polynomials

To derive the recurrence relations for Laguerre polynomials, we use their generating function $G(x,t)$:

$$G(x,t) = \frac{1}{1-t}\exp\left(-\frac{xt}{1-t}\right) = \sum_{n=0}^{\infty} L_n(x)t^n$$

Differentiating $G(x,t)$ with respect to t using the product rule gives:

$$\frac{\partial G}{\partial t} = \frac{\partial}{\partial t}\left[(1-t)^{-1}\right]\exp\left(-\frac{xt}{1-t}\right) + (1-t)^{-1}\frac{\partial}{\partial t}\left[\exp\left(-\frac{xt}{1-t}\right)\right]$$

Evaluating the derivative of the exponent by the quotient rule gives:

$$\frac{\partial}{\partial t}\left(-\frac{xt}{1-t}\right) = \frac{(-x)(1-t) - (-xt)(-1)}{(1-t)^2} = \frac{-x}{(1-t)^2}$$

Substituting this back into the derivative equation gives:

$$\frac{\partial G}{\partial t} = \frac{1}{(1-t)^2}\exp\left(-\frac{xt}{1-t}\right) - \frac{x}{(1-t)^3}\exp\left(-\frac{xt}{1-t}\right)$$

Factoring out $G(x,t) = \frac{1}{1-t}\exp\left(-\frac{xt}{1-t}\right)$ gives:

$$\frac{\partial G}{\partial t} = \left[\frac{1}{1-t} - \frac{x}{(1-t)^2}\right]G(x,t) = \frac{1-t-x}{(1-t)^2}G(x,t)$$

Multiplying both sides of the equation by $(1-t)^2$ gives:

$$(1-2t+t^2)\frac{\partial G}{\partial t} = (1-t-x)G(x,t)$$

Substituting the power series $G(x,t) = \sum_{n=0}^{\infty} L_n(x)t^n$ and its derivative $\frac{\partial G}{\partial t} = \sum_{n=0}^{\infty} nL_n(x)t^{n-1}$ into this equation yields:

$$(1-2t+t^2)\sum_{n=0}^{\infty} nL_n(x)t^{n-1} = (1-t-x)\sum_{n=0}^{\infty} L_n(x)t^n$$

Expanding both sides of the summation expression gives:

$$\sum_{n=0}^{\infty} nL_n(x)t^{n-1} - 2\sum_{n=0}^{\infty} nL_n(x)t^n + \sum_{n=0}^{\infty} nL_n(x)t^{n+1} = (1-x)\sum_{n=0}^{\infty} L_n(x)t^n - \sum_{n=0}^{\infty} L_n(x)t^{n+1}$$

Adjusting indices to collect the coefficient of t^n on both sides: - In the first term, shifting $n \to n+1$ gives $(n+1)L_{n+1}(x)$. - In the third term, shifting $n \to n-1$ gives $(n-1)L_{n-1}(x)$. - On the right-hand side, the second term shifting $n \to n-1$ gives $-L_{n-1}(x)$.

Equating the coefficients of t^n on both sides yields:

$$(n+1)L_{n+1}(x) - 2nL_n(x) + (n-1)L_{n-1}(x) = (1-x)L_n(x) - L_{n-1}(x)$$

Rearranging terms to isolate $(n+1)L_{n+1}(x)$ on the left-hand side gives:

$$(n+1)L_{n+1}(x) = 2nL_n(x) + (1-x)L_n(x) - L_{n-1}(x) - (n-1)L_{n-1}(x)$$

Combining the coefficients of $L_n(x)$ and $L_{n-1}(x)$ produces the recurrence relation:

$$(n+1)L_{n+1}(x) = (2n+1-x)L_n(x) - nL_{n-1}(x)$$

Questions

Short form first (2–3 marks), then long form (5–10 marks). **There is no Assam University question for this unit in the paper bank** — gap. The questions below are topic-matched from other universities' papers (non-AU) or standard syllabus-derived questions, marked as such.

Short-form questions

S1. [standard] State the Laplace equation and explain the separation-of-variables method.

Answer in brief: Sections 1–2 — $\nabla^2 u = 0$; assume $u = X(x)Y(y)$, divide through, and each side equals the same separation constant, giving two ordinary differential equations.

S2. [standard] Write the wave equation for a stretched string and state the form of its solution.

Answer in brief: Section 4 — $\partial^2 y/\partial t^2 = c^2\,\partial^2 y/\partial x^2$ with $c^2 = T/\mu$; the solution is a sum of normal modes $\sin(n\pi x/l)\cos(n\pi ct/l)$.

S3. [standard] What are the normal modes and frequencies of a string of length l fixed at both ends?

Answer in brief: Section 4 — modes $\sin(n\pi x/l)$, $n = 1, 2, 3, \ldots$, with frequencies $f_n = nc/2l$ where $c = \sqrt{T/\mu}$.

S4. [standard] State the diffusion equation.

Answer in brief: Section 6 — $\partial u/\partial t = D\,\nabla^2 u$, where D is the diffusion constant; unlike the wave equation it is first order in time, so disturbances decay instead of oscillating.

S5. [standard] Define Hermite polynomials by Rodrigues' formula and give H_0, H_1, H_2 and H_3.

Answer in brief: Section 7 — $H_n(x) = (-1)^n e^{x^2}\frac{d^n}{dx^n}e^{-x^2}$; $H_0 = 1$, $H_1 = 2x$, $H_2 = 4x^2 - 2$, $H_3 = 8x^3 - 12x$.

S6. [standard] State the recurrence relation 2x H_n(x) = 2n H_{n-1}(x) + H_{n+1}(x).

Answer in brief: Section 7 — the three-term recurrence connecting H_{n+1}, H_n and H_{n-1}, proved from the generating function or Rodrigues' formula.

S7. [standard] State the generating function of Hermite polynomials.

Answer in brief: Section 8 — $e^{2xt-t^2} = \sum_{n=0}^{\infty} H_n(x)\frac{t^n}{n!}$.

S8. [standard] Define Laguerre polynomials and give L_0, L_1 and L_2.

Answer in brief: Section 9 — $L_n(x) = \frac{e^x}{n!}\frac{d^n}{dx^n}(x^n e^{-x})$; $L_0 = 1$, $L_1 = 1 - x$, $L_2 = 1 - 2x + x^2/2$.

S9. [standard] State a recurrence relation for Laguerre polynomials.

Answer in brief: Section 9 — $(n+1)L_{n+1}(x) = (2n + 1 - x)L_n(x) - nL_{n-1}(x)$.

S10. [standard] In which coordinate systems is the Laplace equation solved in this unit, and what special functions appear in the cylindrical and spherical cases?

Answer in brief: Sections 2–3 — rectangular (sine and cosine series), cylindrical (radial functions), spherical (Legendre polynomials in the angular part, powers of r in the radial part).

Long-form questions

L1. [non-AU] Derive the wave equation for the vibration of a string. — Mumbai University, Applied Mathematics-4, Dec 2013, 7 marks.

What to write: Section 4 in full — take a small element of the string, apply Newton's second law in the transverse direction, use the tension and the small-slope approximation, and arrive at $\partial^2 y/\partial t^2 = c^2\,\partial^2 y/\partial x^2$.

L2. [non-AU] Prove the recurrence relation 2x H_n(x) = 2n H_{n-1}(x) + H_{n+1}(x) for Hermite polynomials. — Gauhati University, B.Sc 4th Sem Physics (Major) Paper M-4.1, 2015, Q3d.

What to write: Section 7 in full — differentiate the generating function with respect to t (or use Rodrigues' formula), equate coefficients of t^n, and rearrange to the three-term recurrence.

L3. [non-AU] Show that the generating function for the Hermite polynomial is e^{2xt-t^2} = sum H_n(x) t^n/n! — Gauhati University, B.Sc 4th Sem Physics M-4.1, 2015, Q3b-ii.

What to write: Section 8 in full — expand $e^{2xt-t^2} = e^{-(t-x)^2+x^2}$ as a Taylor series in t and identify the coefficient of $t^n/n!$ with $H_n(x)$ from Rodrigues' formula.

L4. [non-AU] Solve the Laplace equation in two dimensions by separation of variables for a rectangular region, with u(0,y) = u(l,y) = u(x,0) = 0 and u(x,a) = sin(n pi x/l); also state the recurrence relations for Laguerre polynomials and solve the diffusion equation by separation of variables. — Mumbai University, Applied Mathematics-4, Dec 2013, 5 marks (rectangular problem); Laguerre recurrence and the diffusion equation are standard / syllabus-derived, no AU source.

What to write: Section 2 for the rectangular Laplace problem in full (separate, apply the three zero boundary conditions, keep the sine modes, match the top boundary); Section 9 for the Laguerre recurrences; Section 6 for the diffusion equation by separation of variables.

Sources

These notes were written from the unit syllabus only, as the user asked. No external PDF, textbook, or lecture notes were used. The questions are from the Assam University paper bank and other universities' papers (see the Questions section above), not from any source used for the notes.